

A)

```

Snakefile
BIN5506-assignment3 > Snakefile > ...
1  SRA="SRR1972739"
2  REF_ID="AF086833.2"
3  RESULTS_FOLDER="results"
4  RAW_DIR=f"{RESULTS_FOLDER}/raw"
5  ALIGNED_DIR=f"{RESULTS_FOLDER}/aligned"
6  VARIANT_DIR=f"{RESULTS_FOLDER}/variants"
7  ANNOTATED_DIR=f"{RESULTS_FOLDER}/annotated"
8  QC_DIR=f"{RESULTS_FOLDER}/qc"
9  SNPEFF_DIR=f"{RESULTS_FOLDER}/snpeff"
10 SNPEFF_DATA_DIR=f"{SNPEFF_DIR}/data/reference_db"
11 SNAKEMAKE_DIR=f"{RESULTS_FOLDER}/snakemake"
12
13 rule all:
14     input:
15         f"{SNAKEMAKE_DIR}/dirs_created",
16         f"{RAW_DIR}/reference.fasta",
17         f"{RAW_DIR}/{SRA}.fastq",
18         f"{QC_DIR}/{SRA}_fastqc.html",
19         f"{QC_DIR}/{SRA}_fastqc.zip",
20         f"{RAW_DIR}/reference.dict",
21         f"{ALIGNED_DIR}/aligned.sam",
22         f"{ALIGNED_DIR}/aligned_sorted.bam"

```

B)

```

Dockerfile
1  FROM conda/miniconda3
2
3  RUN conda config --set channel_priority strict
4
5  RUN conda config --add channels bioconda
6
7  RUN conda config --prepend channels conda-forge
8
9  RUN conda install entrez-direct sra-tools fastqc samtools bwa gatk4 snpeff snakemake -y
10
11 COPY Snakefile .
12
13 # Default command: run Snakemake
14 CMD snakemake --cores 1

```

C)

```

Total reclaimed space: 9.22GB
(base) ubuntu@ip-172-31-18-0:~/BIN5506-assignment4$ sudo docker build -t abrar_assignment_3 .
[+] Building 1100.0s (12/12) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 364B
=> [internal] load metadata for docker.io/conda/miniconda3:latest
=> [auth] conda/miniconda3:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/6] FROM docker.io/conda/miniconda3:latest@sha256:2141bfc4b60cc5eb332402207c80d884daa72fcef4a48b50f3ccadd934d1f3d03

```

```
(base) ubuntu@ip-172-31-18-0:~/BINF5506-assignment4$ sudo docker run abrar_assignment_3:latest
Annotating variants with snpEff...
Annotated variants with snpEff!
Pipeline completed successfully! Check the folders in results for output files.
[Wed Apr  8 20:52:23 2026]
Finished job 9.
9 of 10 steps (90%) done
Select jobs to execute...

[Wed Apr  8 20:52:23 2026]
localrule all:
  input: results/snakemake/dirs_created, results/raw/reference.fasta, results/raw/SRR1972739.fastq, results/qc/SRR1972739_fastqc.html, results/qc/SRR1972739_fastqc.zip, results/raw/reference.dict, results/aligned/aligned.sam, results/aligned/aligned.sorted.bam, results/aligned/dedup.bam, results/aligned/dup_metrics.txt, results/variants/raw_variants.vcf, results/variants/filtered_variants.vcf, results/snpEff/data/reference_db/genes.gbk, results/snpEff/snpEff.config, results/annotated/annotated_variants.vcf, results/snpEff/snpEff.html
  jobid: 0
  reason: Input files updated by another job: results/raw/reference.dict, results/snpEff/snpEff.config, results/variants/raw_variants.vcf, results/aligned/dedup.bam, results/variants/filtered_variants.vcf, results/qc/SRR1972739_fastqc.html, results/snpEff/snpEff.html, results/snpEff/data/reference_db/genes.gbk, results/aligned/aligned.sam, results/qc/SRR1972739_fastqc.zip, results/aligned/aligned.sorted.bam, results/raw/reference.fasta, results/raw/SRR1972739.fastq, results/aligned/dup_metrics.txt, results/annotated/annotated_variants.vcf, results/snake/dirs_created
  resources: tmpdir=/tmp

[Wed Apr  8 20:52:23 2026]
Finished job 0.
10 of 10 steps (100%) done
Complete log: .snakemake/log/2026-04-08T205130.867016.snakemake.log

(base) ubuntu@ip-172-31-18-0:~/BINF5506-assignment4/results$ tree
.
├── aligned
│   ├── aligned.sam
│   ├── aligned.sorted.bam
│   ├── dedup.bam
│   ├── dedup.bam.bai
│   └── dup_metrics.txt
├── annotated
│   └── annotated_variants.vcf
├── qc
│   ├── SRR1972739_fastqc.html
│   └── SRR1972739_fastqc.zip
├── raw
│   ├── SRR1972739
│   │   └── SRR1972739.sra
│   ├── SRR1972739.fastq
│   ├── reference.dict
│   ├── reference.fasta
│   ├── reference.fasta.amb
│   ├── reference.fasta.ann
│   ├── reference.fasta.bwt
│   ├── reference.fasta.fai
│   ├── reference.fasta.pac
│   └── reference.fasta.sa
├── snakemake
│   └── dirs_created
└── snpEff
    ├── data
    │   └── reference_db
    │       ├── genes.gbk
    │       ├── sequence.bin
    │       └── snpEffectPredictor.bin
```

D)

[Repositories](#) / [abrar_assignment_3](#) / [General](#)

nobody1962/abrar_assignment_3

Last pushed 1 minute ago

Repository size: 2.2 GB

☆ 0

↓ 0

Add a description

Add a category

Docker commands

To push a new tag to this repository:

```
docker push nobody1962/abrar_assignment_3:tagname
```

Public view

General

Tags

Image Management

Collaborators

Webhooks

Settings

Tags

DOCKER SCOUT INACTIVE

Activate

This repository contains 0 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	1 minute

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-

[illegible]

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 0313-0511-0421

voclabs/user4544613-n10005099@humber.ca

AWS Batch

Create Batch resources

Step 1

Configure job and orchestration type

Step 2

Create a compute environment

Step 3

Create a job queue

Step 4

Create a job definition

Step 5

Create a job

Step 6

Review and create

Create a job queue

Jobs are submitted to a job queue, where they reside until they are able to be scheduled to run in a compute environment.

Job queue configuration

Name

Lab4_BatchPIPELINE

Job queue name must be 1-128 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_). Name must be unique.

Priority

1

Job queues with a higher value for priority are given preference for job scheduling in environments.
Enter a whole number between 0-1000.

Compute environment order

Compute environment added automatically

The "getting-started-fargate-compute-env" compute environment created in the previous step will automatically be added to the this queue.

Cancel

Previous

Next

AWS Batch

Job queues

Lab4_BatchPIPELINE

Lab4_BatchPIPELINE

Info

Last updated: Now

Actions

View jobs

Status and usage

State

Enabled

Status

Valid - JobQueue Healthy

Status reason

JobQueue Healthy

Total capacity utilization snapshot

2 vCPUs

Details

ARN

arn:aws:batch:us-east-1:031305110421:job-queue/Lab4_BatchPIPELINE

Job queue type

ECS Fargate

Priority

0

Scheduling algorithm

First-in, First-out (FIFO)

Connected compute environments

Order	Name	Status	State	Maximum vCPUs
1	Assignment3_BatchPIPELINE	Valid	Enabled	256

Job queue snapshot

Tags

JSON

Job queue snapshot (1)

F)

AWS Batch

Create Batch resources

Enable if you want the elastic network interface attached to the Fargate job to be assigned a public IP address. This is required if your job needs outbound network access.

Ephemeral storage

The amount of ephemeral storage to allocate for the task. Defaults to 20 GiB when no value is provided.

50

Can only be used with platform version 1.4.0 or later. Must be a value between 21 GiB and 200 GiB.

Execution role

LabRole

Create an execution role

Container configuration

Image

nobody1962/abrar_assignment_3

The image used to start a container. This string is passed directly to the Docker daemon. Images in the Docker Hub registry are available by default. You can also specify other repositories with repository-url/image:tag.
Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.

Command - optional

The command you want to execute when your job is run.

[illegible]

G)

AWS Batch > Dashboard

Account ID: 0313-0511-0421
vocalabs/user4544613@m10005099@humber.ca

Introducing Quota Management
AWS Batch now supports quota management, enabling administrators to allocate shared compute resources across teams and projects through quota shares with capacity limits, resource-sharing strategies, and priority-based preemption - currently available for SageMaker Training job queues. Let us know what you think.

Notifications: 0 0 0 0 2 0

Reset to default layout Add widgets

Auto-refresh Dashboard last updated: 05:51:36 PM. Auto-refreshes every 60 seconds.

Jobs overview

Runnable	Running	Succeeded	Failed
0	0	1	0

Job queue overview

Find job queue

Job queue	Total capacity utilization snapshot	Priority	Submitted	Pending	Runnable	Starting	Running	Succeeded
getting-started-fargate-job-queue	-	0	0	0	0	0	0	0
Lab4_BatchPIPELINE	-	0	0	0	0	0	0	1

AWS Batch > Jobs > Lab4_BatchPIPELINE

Keywords: Max results: Sorting: Asc Filter logs Reset

Logs can be filtered based on keywords, max results, and sort order. To remove applied filters, select "Reset" and then "Filter logs".

Investigate with AI - new 2026-04-08T21:41:23 > 2026-04-08T21:53:39 UTC timezone Explore related

Lab4_BatchPIPELINE/default/89ac022e5cfc455d92ebadaad9e1142c

#	@timestamp	@message
1	2026-04-08T21:43:20.569Z	Building DAG of jobs...
2	2026-04-08T21:43:20.582Z	Using shell: /bin/bash
3	2026-04-08T21:43:20.582Z	Rules claiming more threads will be scaled down.
4	2026-04-08T21:43:20.582Z	Provided cores: 1 (use --cores to define parallelism)
5	2026-04-08T21:43:20.587Z	Select jobs to execute...
6	2026-04-08T21:43:20.587Z	total 10 1 1
7	2026-04-08T21:43:20.587Z	all 1 1 1
8	2026-04-08T21:43:20.587Z	download_fastq 1 1 1
9	2026-04-08T21:43:20.587Z	Job stats:
10	2026-04-08T21:43:20.587Z	create_dirs 1 1 1
11	2026-04-08T21:43:20.587Z	Variant_Calling 1 1 1
12	2026-04-08T21:43:20.587Z	snpEff_Annotation 1 1 1
13	2026-04-08T21:43:20.587Z	Create_BAM 1 1 1
14	2026-04-08T21:43:20.587Z	QC_check 1 1 1
15	2026-04-08T21:43:20.587Z	Create_SAM 1 1 1
16	2026-04-08T21:43:20.587Z	snpEff_database 1 1 1
17	2026-04-08T21:43:20.587Z	download_fastq 1 1 1
18	2026-04-08T21:43:20.587Z	
19	2026-04-08T21:43:20.587Z	job count min threads max threads
20	2026-04-08T21:43:20.590Z	jobid: 1
21	2026-04-08T21:43:20.590Z	rule create_dirs:

AWS Batch > Jobs > Lab4_BatchPIPELINE

Account ID: 0313-0511-0421
vocalabs/user4544613@m10005099@humber.ca

Last updated: Now Clone Cancel Terminate

Status and capacity usage

Status: Succeeded

Status reason: Essential container in task exited

Capacity usage: 2 vCPUs

Job details

ID: 688e4325-94ca-4a7f-bdcd-b0d2fb038f3	ARN: arnaws:batch:us-east-1:031305110421:job/688e4325-94ca-4a7f-bdcd-b0d2fb038f3	Created at: Apr 08, 2026, 21:41:23 UTC+00:00
Started at: Apr 08, 2026, 21:43:19 UTC+00:00	Stopped at: Apr 08, 2026, 21:44:22 UTC+00:00	Log stream name: Lab4_BatchPIPELINE/default/89ac022e5cfc455d92ebadaad9e1142c

Job attempts Job configuration Containers Logging Consumable resources Tags JSON

Job attempts

Created at	Started at	Stopped at
Apr 08, 2026, 17:41:23	Apr 08, 2026, 17:43:19	Apr 08, 2026, 17:44:22